

# IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 17: The "Blob" Hunt (FAST Detector Game)

## ACTIVITY CONCEPT & GAME RULES

**Concept:** The FAST (Features from Accelerated Segment Test) algorithm evaluates a candidate pixel  $p$  by considering a Bresenham circle of 16 pixels surrounding it. Candidate pixel  $p$  with intensity  $I_p$  is classified as a corner if there exists a set of  $N$  contiguous pixels in the circle that are all strictly brighter than  $I_p + t$  or darker than  $I_p - t$  (where  $t$  is a threshold value).

| Parameter                        | Standard Value / Condition             | Game Criterion                                                     |
|----------------------------------|----------------------------------------|--------------------------------------------------------------------|
| Center Pixel Intensity ( $I_p$ ) | Value between 1 and 10                 | Reference point for comparative intensity threshold.               |
| Circle Radius / Perimeter        | 16 surrounding pixels                  | Bresenham circle of radius 3 around pixel $p$ .                    |
| Contiguous Threshold ( $N$ )     | FAST-9 ( $N=9$ ) or FAST-12 ( $N=12$ ) | At least 9 contiguous pixels must be brighter/darker.              |
| High-Speed Test (Optimization)   | Check pixels 1, 5, 9, 13 first         | At least 3 of these 4 pixels must satisfy the threshold condition. |

## WORKSHEET QUESTIONS & SOLUTIONS

1. 1. How does FAST achieve significantly high computational speed compared to Harris corner detection?

FAST performs a high-speed exclusion test by checking only 4 cardinal pixels (pixels 1, 9, 5, and 13 on the 16-pixel circle). If  $p$  is a corner, at least 3 of these 4 pixels must be brighter than  $I_p + t$  or darker than  $I_p - t$ . If this condition is not met, candidate  $p$  is immediately rejected without calculating image gradients or multi-directional derivatives.

2. 2. What is the role of the threshold parameter  $t$  in FAST corner detection?

The intensity threshold  $t$  controls detector sensitivity. It prevents false corner detections due to minor noise or smooth lighting gradients by requiring that surrounding circle pixels differ significantly from the center pixel's intensity ( $I_{\text{pixel}} > I_p + t$  or  $I_{\text{pixel}} < I_p - t$ ).